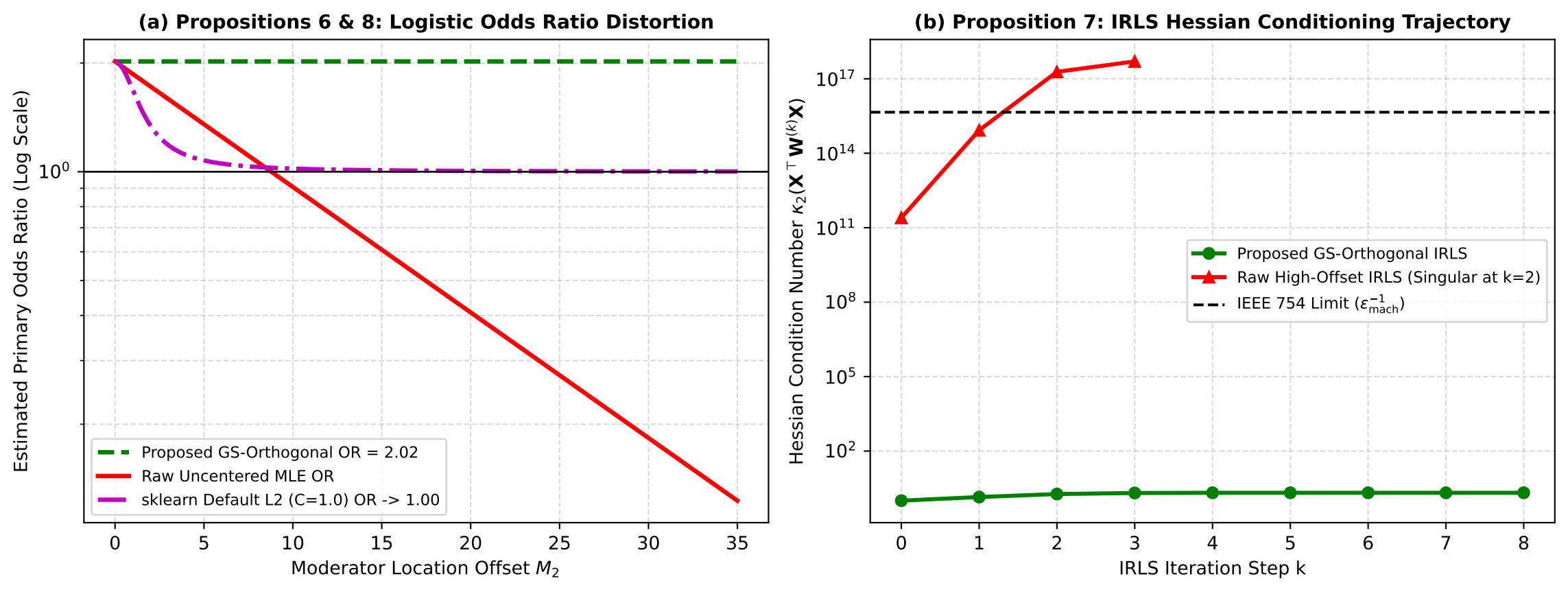
(a) Propositions 6 & 8: Logistic Odds Ratio Distortion
Estimated Primary Odds Ratio (Log Scale)
10^0
Proposed GS-Orthogonal OR = 2.02
Raw Uncentered MLE OR
sklearn Default L2 (C=1.0) OR -> 1.00
0
5
10
15
20
25
30
35
Moderator Location Offset M_2
(b) Proposition 7: IRLS Hessian Conditioning Trajectory
Hessian Condition Number $\kappa_2(\mathbf{X}^\top \mathbf{W}^{(k)} \mathbf{X})$
10^{17}
10^{14}
10^{11}
10^8
10^5
10^2
Proposed GS-Orthogonal IRLS
Raw High-Offset IRLS (Singular at k=2)
IEEE 754 Limit ($\epsilon_{\text{mach}}^{-1}$)
0
1
2
3
4
5
6
7
8
IRLS Iteration Step k